MATH 310/STAT 410 MIDTERM 1

FALL 2024

Problem 1

Solution.

(a) Let Y be a random variable such that if either A or B is selected, then $Y = 1$, and otherwise $Y = 0$. The distribution of Y is as follows:

$$Y = \begin{cases} 1, & \text{with probability } 2/3 \\ 0, & \text{with probability } 1/3 \end{cases}$$

Conditional on $Y = 1$, X has the same distribution as $T_{AB} + T_C$, and conditional on $Y = 0$, $X = T_C$. Thus

$$\mathbb{E}(X|Y = 1) = \mathbb{E}(T_{AB} + T_C) = 12 + 9 = 21 \tag{1}$$

$$\mathbb{E}(X|Y = 0) = \mathbb{E}(T_C) = 9, \tag{2}$$

and it follows that

$$\mathbb{E}(X) = \mathbb{E}[\mathbb{E}(X|Y)] = \frac{2}{3} \cdot 21 + \frac{1}{3} \cdot 9 = 14 + 3 = \boxed{17}.$$

(b) From (1) and (2), we have

$$\mathrm{Var}[\mathbb{E}(X|Y)] = \frac{2}{3}(21 - 17)^2 + \frac{1}{3}(9 - 17)^2 = \frac{2 \cdot 4^2 + 1 \cdot 8^2}{3} = 32.$$

Similarly, conditional on the value of Y, we have the variance of X as

$$\mathrm{Var}(X|Y = 1) = \mathrm{Var}(T_{AB} + T_C) = 4 + 3 = 7$$

$$\mathrm{Var}(X|Y = 0) = \mathrm{Var}(T_C) = 3,$$

which yields $\mathbb{E}[\mathrm{Var}(X|Y)] = \frac{2}{3} \cdot 7 + \frac{1}{3} \cdot 3 = \frac{17}{3}$. Putting the two terms in the law of total variance together, we get

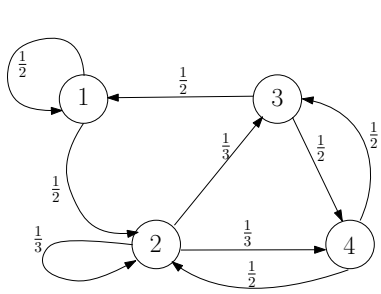
$$\mathrm{Var}(X) = \mathrm{Var}[\mathbb{E}(X|Y)] + \mathbb{E}[\mathrm{Var}(X|Y)] = 32 + \frac{17}{3} = \boxed{\frac{113}{3}}.$$

□

Problem 2. (30 pts)

Solution.

(a).



The transition graph is shown on the left. There is a single communication class $\{1, 2, 3, 4\}$, which is recurrent.

The class has period 1.

(b). The probability is given by

$$\begin{aligned}
&P(X_5 = 2, X_3 = 2 | X_0 = 3) = P(X_3 = 2 | X_0 = 3) \cdot P(X_2 = 2 | X_0 = 2) \\
&= [P(3 \to 1 \to 1 \to 2) + P(3 \to 1 \to 2 \to 2) + P(3 \to 4 \to 2 \to 2)] \\
&\quad \cdot [P(2 \to 2 \to 2) + P(2 \to 4 \to 2)] \\
&= \left[\left(\frac{1}{2}\right)^3 + \left(\frac{1}{2}\right)^2 \cdot \frac{1}{3} + \left(\frac{1}{2}\right)^2 \cdot \frac{1}{3}\right] \cdot \left[\left(\frac{1}{3}\right)^2 + \frac{1}{3} \cdot \frac{1}{2}\right] \\
&= \frac{7}{24} \cdot \frac{5}{18}
\end{aligned}$$

(c). Denote the probability by p. By law of total probability, we have

$$p = \frac{1}{2} + \frac{1}{2} P(T_1 < T_2 | X_0 = 4) = \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} p \implies \boxed{p = \frac{2}{3}}.$$

(d). There is a limiting distribution, because the transition probability matrix is regular. To see this, we note that for any pair of states (i, j), there is a path of length 3 starting from i and ending at j such that every step on the path occurs with positive probability.

□

Problem 3. (25 pts)

Solution.

(a) The process is a Markov chain because given $X_n = x$ where $x = 1, 2$ or 3, the number of working light bulbs on the $(n+1)$-th day will be exclusively determined by the x working light bulbs on the n-th day, and the number of working light bulbs on the days before the n-th day is irrelevant.

(b) The transition probability matrix is as follows, with the state space $\mathcal{S} = \{1, 2, 3\}$.

$$\mathbf{P} = \begin{array}{c} \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{ccc} 1 & 2 & 3 \end{array} \\ \begin{pmatrix} 1-p & 0 & p \\ 2p(1-p) & (1-p)^2 & p^2 \\ 3p^2(1-p) & 3p(1-p)^2 & p^3 + (1-p)^3 \end{pmatrix} \end{array} = \begin{array}{c} \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{ccc} 1 & 2 & 3 \end{array} \\ \begin{pmatrix} q & 0 & p \\ 2pq & q^2 & p^2 \\ 3p^2q & 3pq^2 & p^3 + q^3 \end{pmatrix} \end{array}$$

The Markov chain is irreducible.

(c) Suppose 3 new light bulbs were replaced just before day 0, and the Markov chain starts with $X_0 = 3$. Let $T_3 \stackrel{\text{def}}{=} \min\{n > 0 : X_n = 3\}$ be the first time (after day 0) such that there are 3 working light bulbs. Let T be the first time (after day 0) such that there are 3 working light bulbs, **and right on the night before, 3 light bulbs are replaced**. Note that these are two different random variables, and we were asked to compute $\mathbb{E}(T | X_0 = 3)$. By law of total expectation, we have

$$\begin{aligned}
\mathbb{E}(T_3 | X_0 = 3) &= 3p^2q[1 + \mathbb{E}(T_3 | X_0 = 1)] + 3pq^2[1 + \mathbb{E}(T_3 | X_0 = 2)] + (p^3 + q^3) \cdot 1 \\
\mathbb{E}(T | X_0 = 3) &= 3p^2q[1 + \mathbb{E}(T | X_0 = 1)] + 3pq^2[1 + \mathbb{E}(T | X_0 = 2)] \\
&\quad + p^3 \cdot 1 + q^3[1 + \mathbb{E}(T | X_0 = 3)]
\end{aligned}$$

Since when starting from $X_0 = 1$ or $X_0 = 2$, state 3 must follow the light bulb replacement, we must have

$$\mathbb{E}(T_3 | X_0 = 1) = \mathbb{E}(T | X_0 = 1), \quad \text{and } \mathbb{E}(T_3 | X_0 = 2) = \mathbb{E}(T | X_0 = 2).$$

Subtracting the two equations, we get

$$(1 - q^3)\mathbb{E}(T | X_0 = 3) = \mathbb{E}(T_3 | X_0 = 3) \stackrel{\text{def}}{=} \mu_3.$$

We compute $\mu_3 \stackrel{\text{def}}{=} \mathbb{E}(T_3|X_0 = 3)$ using the relation that $\mu_3 = \frac{1}{\pi_3}$ (Theorem 3.6). The equation $\boldsymbol{\pi}\mathbf{P} = \boldsymbol{\pi}$ gives

$$\begin{cases} q\pi_1 + 2pq\pi_2 + 3p^2q\pi_3 = \pi_1 \\ q^2\pi_2 + 3pq^2\pi_3 = \pi_2 \\ p\pi_1 + p^2\pi_2 + (p^3 + q^3)\pi_3 = \pi_3 \\ \pi_1 + \pi_2 + \pi_3 = 1 \end{cases}$$

We solve and get $\pi_3 = \frac{1+q}{3q^3+3q^2+4q+1} \approx 0.2061$, and thus $\mu_3 = \frac{1}{\pi_3} \approx 4.851$. It follows that

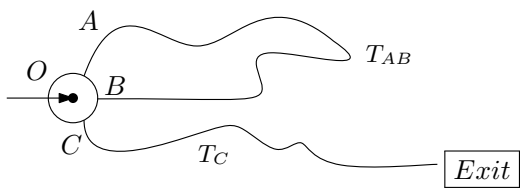
$$\mathbb{E}(T|X_0 = 3) = \frac{\mu_3}{1 - q^3} \approx \boxed{17.90}.$$

$\square$

MATH 310/STAT 410 MIDTERM 1

FALL 2024

Problem 1. (25 pts) Suppose you have lost the trail map and find yourself at an intersection O with three possible routes ahead: A, B, and C, where A and B form a loop and C leads you to the exit of the trail. The travel times needed to complete the loop and route C are **independent** random variables, with mean and variance given below.



$$\mathbb{E}(T_{AB}) = 12, \mathrm{Var}(T_{AB}) = 4$$

$$\mathbb{E}(T_C) = 9, \quad \mathrm{Var}(T_C) = 3.$$

Without know which routes A, B, C are, your strategy to get to the exit is the following. Firstly, you pick one route from A, B, C **at random** equally likely, (i.e., with probability 1/3 each) and follow the route. If you have picked route C, then you will get to the exit successfully in just one attempt. Otherwise, if the route takes you back to the intersection O, then you know you must have chosen either A or B. Thus, upon returning to O after the first attempt, you will know which one is route C and will then follow it to the exit of the trail in your second attempt.
Let X denote the total time you spend in order to get to the exit of the trail.

(a). (10 pts) Find $\mathbb{E}(X)$.

(b). (15 pts) Find $\mathrm{Var}(X)$.

Problem 2. (30 pts) Consider the following Markov chain with state space $\{1, 2, 3, 4\}$.

$$\mathbf{P} = \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \end{array} \begin{array}{c} \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} \\ 0 & \frac{1}{2} & \frac{1}{2} & 0 \end{pmatrix} \end{array}$$

(a). (10 pts) Draw the transition graph for this Markov chain. Find the communication class(es). Identify the **period** and the **recurrence/transience** property for each communication class.

(b). (8 pts) Find $P(X_8 = 2, X_6 = 2 | X_3 = 3)$. No need to do matrix-matrix multiplication and you don't have to evaluate the numerical value of fractions.

(c). (5 pts) For $i = 1, 2, 3, 4$, let $T_i \stackrel{\text{def}}{=} \min\{n \geq 1 : X_n = i\}$ be the first time to visit state i.
Find $P(T_1 < T_2 | X_0 = 3)$, i.e., the probability of visiting state 1 before state 2, if the Markov chain is started from state 3.

(d). (2 + 5 pts) Is there a limiting distribution? Explain why.

Problem 3. (25 pts) A professor has three light bulbs in his garage. Suppose (i) each light bulb burns out during a day with probability $p = 0.1$, independent of the other light bulbs, and (ii) when **all three light bulbs** are burned out, they are replaced together at the end of the day, and the next day **starts with <u>three</u> working light bulbs**. Let X_n denote the number of working light bulbs at the beginning of the n-th day.

(a) (5 pts) Explain why $(X_n)_{n\geq 0}$ is a Markov chain.

(b) (10 pts) Write down the <u>state space</u> and <u>the transition probability matrix</u> of $(X_n)_{n\geq 0}$. Is the Markov chain irreducible? (For the ease of computation, you *may* leave p and $q \stackrel{\text{def}}{=} 1-p$ in the transition probabilities).

(c) (10 pts) What is the expected time between light bulb replacements?

Formula Sheet

DISCRETE DISTRIBUTIONS	
Distribution	**PMF, Expectation, Variance**
Uniform $(1, \dots, n)$	$P(X=k)=\frac{1}{n}, \quad k=1, \dots, n$ $E(X)=\frac{n+1}{2} \quad Var(X)=\frac{n^2-1}{12}$
Binomial	$P(X=k)=\binom{n}{k}p^k(1-p)^{n-k}, \quad k=0,1, \dots, n$ $E(X)=np \quad Var(X)=np(1-p)$
Poisson	$P(X=k)=\frac{e^{-\lambda}\lambda^k}{k!}, \quad k=0,1,2, \dots$ $E(X)=\lambda \quad Var(X)=\lambda$
Geometric	$P(X=k)=(1-p)^{k-1}p, \quad k=1,2, \dots$ $E(X)=\frac{1}{p} \quad Var(X)=\frac{1-p}{p^2}$ $P(X>t)=(1-p)^t$
Negative binomial	$P(X=k)=\binom{k-1}{r-1}p^r(1-p)^{k-r},$ $k=r, r+1, \dots, \quad r=1,2, \dots$ $E(X)=\frac{r}{p} \quad Var(X)=\frac{r(1-p)}{p^2}$
Hypergeometric	$P(X=k)=\frac{\binom{D}{k}\binom{N-D}{n-k}}{\binom{N}{n}}, k=0,1, \dots, n$ $E(X)=\frac{nD}{N} \quad Var(X)=\frac{nD(N-D)}{N^2}\left(1-\frac{n-1}{N-1}\right)$

CONTINUOUS DISTRIBUTIONS	
Distribution	**Density, CDF, Expectation, Variance**
Uniform (a,b)	$f(x)=\frac{1}{b-a}, \quad a<x<b$ $F(x)=\frac{x-a}{b-a}, \quad a<x<b$ $E(X)=\frac{a+b}{2} \quad Var(X)=\frac{(b-a)^2}{12}$
Exponential	$f(x)=\lambda e^{-\lambda x}, \quad x>0$ $F(x)=1-e^{-\lambda x}, \quad x>0$ $E(X)=\frac{1}{\lambda} \quad Var(X)=\frac{1}{\lambda^2}$
Normal	$f(x)=\frac{1}{\sigma\sqrt{2\pi}}\exp\left(\frac{-(x-\mu)^2}{2\sigma^2}\right), \quad -\infty<x<\infty$ $E(X)=\mu \quad Var(X)=\sigma^2$
Gamma	$f(x)=\lambda e^{-\lambda x}\frac{(\lambda x)^{r-1}}{\Gamma(r)}, \quad x>0,$ where $\Gamma(r)=\int_0^\infty t^{r-1}e^{-t}\,dt$ $E(X)=\frac{r}{\lambda} \quad Var(X)=\frac{r}{\lambda^2}$
Beta	$f(x)=\frac{1}{B(\alpha,\beta)}x^{\alpha-1}(1-x)^{\beta-1}, \quad 0<x<1,$ where $B(\alpha,\beta)=\frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}.$ $E(X)=\frac{\alpha}{\alpha+\beta} \quad Var(X)=\frac{\alpha\beta}{(\alpha+\beta+1)(\alpha+\beta)^2}$

- If $X=\begin{bmatrix} a & b \\ c & d \end{bmatrix}$, then $X^{-1}=\frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$
- $1+x+x^2+\cdots=\frac{1}{1-x}$, if $|x|<1$.
- If $b^2-4ac\geq 0$, then the roots of the quadratic equation $ax^2+bx+c=0$ are $x_{1,2}=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$.
- Nonstandard integration formula:

  Integration by part: $\int x^n e^{-x}=-x^n e^{-x}+n\int x^{n-1}e^{-x}dx$

Name _____ Page _____

Problem 1

Part (a)

Part (b)

Name _____ Page _____

Problem 2

Part (a)

Part (b)

Name _____ Page _____

Problem 2

Part (c)

Part (d)

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Problem 3

Part (a)

Part (b)

Name _____ Page _____

Problem 3

Part (c)